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Elective Egg Freezing for Non-Medical Reasons (Scientific Impact Paper No. 63, 2026 Second Edition)

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Plain Language Summary

The average age of childbirth in the UK in 2022 was 30.9 years, having increased by over 3 years in the past 2 decades, and 4 years since 1980. Although a woman's fertility declines markedly in her late-30s and early-40s, more women are now starting their family at this stage of their lives. Related to this, increasing numbers of women are storing their eggs (oocytes) to give them the potential opportunity to have a baby in the future. The number of egg freezing cycles accounts for 5% of in vitro fertilisation (IVF) cycles but the number of cycles using stored eggs is much lower. The use of a technique of rapid freezing called vitrification gives success rates almost as good as using fresh eggs. The reasons behind the increase in egg freezing are complex, but the most common reason given by women storing eggs is that they do not have a partner and are concerned that by the time they do find themselves in a relationship within which they wish to start a family, they may not be able to do so. This paper addresses the growing use of egg freezing and how it may have been promoted.

It is essential that women are very clearly informed about the likely success rates of egg freezing, particularly as it is primarily provided by the private sector, with the associated concerns of financial costs and inappropriate or inaccurate marketing. Its success is strongly dependent on the age of the woman at the time of freezing her eggs, with much higher success rates in those under the age of 35 years. The relevant legislation changed in 2022 and now allows women to store eggs for 55 years, although consent for this needs to be renewed every 10 years. This now supports the better success rates when women store eggs at a younger age. We conclude that elective egg freezing provides women with an opportunity to act on the age-related drop in their fertility. At present, most women who are doing this are already in their later 30s when the success rates are limited. We strongly support the need for improved and continuing education of both women and men regarding the decline in female fertility with age.

1 | Background

The age at which women have their children has been increasing steadily for many years in developed countries. For example, the number of births to women over 35 years in the UK has tripled since 1980 [1]. Although in the past few years there has been a reduction in births in women in the 30–34 and 35–40 age groups, the birth rate in women over 40 is still increasing (Figure 1) [1].

Societal changes have impacted the time spent in education as well as in the pursuit of financial independence and partnerships that may lead to parenthood. However, age is the key determinant of female fertility, as the population of non-growing primordial follicles in the ovaries is established during fetal life and declines progressively until the menopause. In addition, the inability to maintain chromosomal integrity results in a decline in oocyte quality. This double jeopardy results in a fall in the likelihood of conceiving with age, along with an increase in the risk of such a pregnancy resulting in miscarriage.

There is clear evidence that provision of fertility education increases fertility awareness about fecundity, infertility, risk factors for reduced fertility, and reproductive technologies [2] and that fertility education can advance the timing of births in women with partners [3]. Men have similar parenthood aspirations as women, but may have limited knowledge of the impact of age on female fertility [4].

There is increasing awareness of the need to improve public knowledge and understanding of how fertility changes with age, in addition to other aspects of family planning and what fertiity treatments can and cannot achieve, as highlighted by the International Reproductive Health Education Collboration (https://www.eshre.eu/irhec) and the Fertility Educational Initiative of the British Fertility Society (https://www.britishfertilitysociety.org.uk/fei).

The divergence between women's reproductive ambitions and oocyte biology has led to increasing numbers of women utilising reproductive technology to undergo ovarian stimulation followed by the recovery and cryopreservation of oocytes (egg freezing) to allow deferment of their reproductive potential. These oocytes can then be stored and used if more conventional ways of starting a family do not occur. This procedure is widely known as 'social egg freezing', but the term 'elective egg freezing' does not have judgemental overtones and is preferred. The UK legislation behind this changed in 2022 and now allows egg freezing for a maximum of 55 years (from only 10 years previously), with a requirement for renewal of consent every 10 years (https://www.hfea.gov.uk/treatments/fertility-preservation/egg-freezing/).

Egg freezing is also used for fertility preservation for purely medical reasons, e.g., for women with a new cancer diagnosis facing gonadotoxic (i.e., potentially sterilising) therapy. There may be a combination of social and medical reasons, such as family history of premature menopause or surgical treatment for severe endometriosis, which can lead to earlier fertility decline. It appears that more women around the world are currently storing eggs for elective rather than medical reasons, and many are already in their late 30s [5] when the efficacy of the procedure is declining. There is, however, a lack of good data on the reasons for egg freezing as it is frequently not recorded. In the UK, the Human Fertilisation and Embryology Authority (HFEA—the UK's independent regulator of fertility treatment) does not record the reason for egg freezing.

This paper addresses the growing use of egg freezing, its efficacy and safety, and ethical considerations.

This guidance is for healthcare professionals who care for women, non-binary and trans people. Within this document we use the terms woman and women's health. However, it is important to acknowledge that it is not only women for whom it is necessary to access women's health and reproductive services in order to maintain their gynaecological health and reproductive wellbeing. Gynaecological and obstetric services and delivery of care must therefore be appropriate, inclusive and sensitive to the needs of those individuals whose gender identity does not align with the sex they were assigned at birth.

2 | Efficacy of Freezing

Freezing to preserve reproductive potential was first achieved with semen storage, and successful pregnancies have been reported using sperm frozen for more than 20 years. Embryo storage followed closely on from in vitro fertilisation (IVF) and is in widespread use. Freezing of eggs proved more technically challenging, reflecting the size of the egg and probably also chromosomal stability as mature eggs are part way through meiosis and thus prone to damage from ice cystals [6]. Despite the first report of a live birth in 1986, pregnancy rates remained low until the development of vitrification (ultra rapid freezing), which transformed success rates and replaced 'slow freezing' (i.e., over several hours) in most IVF laboratories. The American Society for Reproductive Medicine (ASRM) changed the status of oocyte vitrification and warming from 'experimental' to 'established' in 2013 [7]. In an experienced clinic, a vitrified oocyte has the same developmental potential as a fresh oocyte, as it is not subject to age-related decline [8]. The clinical pregnancy rates reported in randomised series using warmed eggs fertilised in vitro are equivalent to fresh IVF treatment [8]. However, these data are from egg donors, who are selected for optimum fertility and are usually much younger than the recipients, and reflect the expertise of centres with greater and longer term experience of vitrification. Much larger case-series reporting experience of elective egg freezing and subsequent use over long periods of time are now being published providing evidence of good success rates [5], although centre-specific data are not widely available. In this UK study of 3328 elective vitrification cycles in 2280 patients, the return rate to use oocytes was 14% with a 36.4% live birth rate [5].

Between 2016 and 2021, the number of egg freezing cycles in the UK trebled from 1362 to 4215 (Figure 2). The most recent data from HFEA show that elective egg storage is the fastest growing treatment in the UK, with 4647 cycles accounting for 5% of all cycles in 2022 [9]. Larger data sets are reported from the US [10], with 38 126 cycles for oocyte banking in 2023, although these data do not distinguish between different indications for egg freezing.

The likelihood of future live birth is dependent on the woman's age at the time of oocyte storage, as well as the number of eggs stored. The number of eggs that are likely to be collected can be indicated by assessing ovarian reserve using follicle stimulating hormone, anti-Müllerian hormone and antral follicle count measurements, often in combination. European data [11] indicate high (more than 90%) cumulative live birth rates in women who had electively frozen eggs at 35 years and under, although that required utilising 24 eggs. This demonstrates that increasing the likelihood of a live birth will often require more than one cycle of oocyte storage, even in younger women. Storage of 10 eggs gave a cumulative live birth rate of 42.8% compared with 25.2% in women aged 36 years and over at the time of storage [11]. Such analyses may represent a 'best case' scenario, which individual centres with less experience may be unable to match. In the UK in 2021, 28% of women storing eggs were 38 or older. Only 41% of women were under 35 years old, although how many were egg donors, who must be aged below 36, is not recorded by the HFEA. Although the technology allows indefinite storage without deterioration, it is very positive that the current UK legal limit for duration of elective oocyte freezing has now been increased to 55 years from 10 years, now supporting the interests of women wishing to freeze eggs at a younger, more effective age.

A cost–benefit analysis compared elective egg freezing with IVF at a later age, and showed that this is a cost-effective approach [12]. While this is based on the key assumption that subsequent

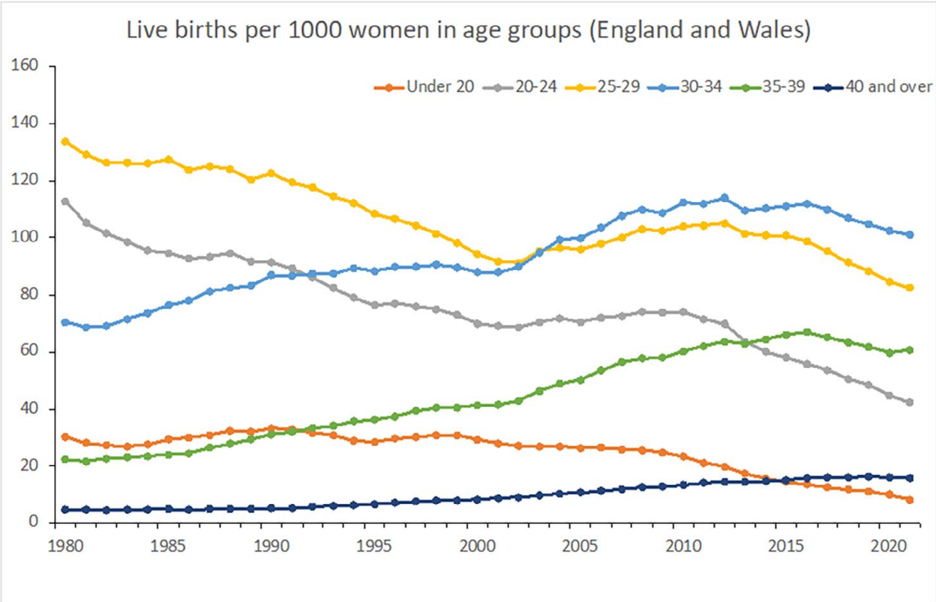

FIGURE 1 | Births in England and Wales 1980–2022 by women's age [1].

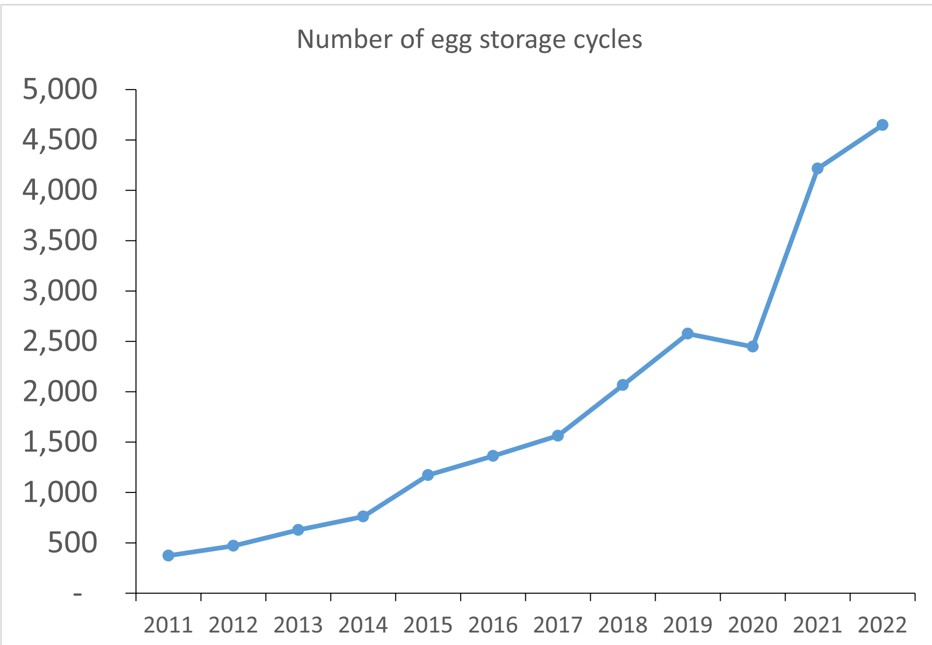

FIGURE 2 | Number of egg freezing cycles in the UK, based on Human Fertilisation and Embryology Authority (HFEA) data 2010–2022 [9].

IVF would be needed—and the much higher costs in the US over the UK—it showed that to have either one or two children, elective egg freezing was more cost effective across a wide age range, with a maximal difference when carried out before age 31.

Freezing of ovarian tissue is also used for medical fertility preservation and could potentially be used for non-medical reasons. It would allow restoration of endocrine function as well as fertility, but the need for surgical intervention (both to remove tissue and later replace it) is an important consideration and there are no data from either medical or elective ovarian tissue transplantation showing long-term hormone-related benefits [13]. The potential benefit of this approach to delaying the menopause has been modelled [14] but there are no actual data on this or the prolongation of fertility. In practice, elective ovarian tissue freezing is not widely available and, in the UK, would require licensing by the Human Tissue Authority.

3 | The Context of Elective Egg Freezing

Treatment of single women in fertility clinics has been legal in the UK since 2008. As described above, the numbers of women freezing eggs for non-medical reasons is increasing and we are beginning to understand what is motivating individuals to make this choice. In addition to the well-recognised demographic forces associated with delaying having a baby, such as education and opportunity in the workplace, rising costs of childbearing, housing factors, workplace inflexibility and the potential wish of men to defer parenthood also contribute to decision making [4]. The lack of a current partner is cited by women undergoing elective egg freezing as the most common reason for postponing childbearing [15]. Studies of these women, in the UK and elsewhere, show that the majority are university educated, in professional employment, between the ages of 36 and 40 years and not in a relationship at the time of egg freezing [15]. They express the desire to have a baby, ideally when in a committed relationship with a partner and with their own eggs, but many would also consider the use of donor sperm if they were unable to find such a partner. Studies of women who have chosen single motherhood through assisted reproduction ('solo mothers'), generally as a second choice due to lack of a partner and thus likely to be a comparable group, show high levels of social support [16, 17]. This contrasts strongly with the negative stereotype of 'single mothers', who are generally characterised by social and economic disadvantage for themselves and their children. While long-term follow-up studies of the resulting children are limited, analyses at up to 8–10 years of age show no concerns about emotional or behavioural development [17].

4 | Ethical and Social Considerations

The ethics of elective egg freezing have attracted much commentary, focusing on issues such as the medicalisation and commodification of reproduction, women's autonomy and discourse about the right time to have a baby. It is likely that many women electively freezing eggs will never return to use them; concerns have therefore been expressed about the potential number of unnecessary medical interventions and the exploitation of reproductive anxiety. There is an additional concern that having eggs in storage might give women a false sense of security in the technology, encouraging them to delay parenthood even longer, with no guarantee of a future pregnancy.

Elective egg freezing has been compared with autologous blood storage for elective surgery [18]. In both situations:

- There is storage of tissue to treat possible future health issues.
- The person is healthy at the point of intervention.
- The procedures are established, with medical and psychological advantages for the patient.
- The need for donor material is avoided.
- There is no certainty the tissue will ever be used.

All indications for oocyte freezing should be evaluated using standard ethical perspectives, such as focusing on the balance between benefit and risk/cost, whether women are concerned about the threat to their future fertility from, for example, chemotherapy or solely increasing age.

Both the Ethics Committee of the ASRM and the Nuffield Council on Bioethics have found elective egg freezing (termed 'planned oocyte cryopreservation' by the ASRM) to be ethically permissible, using as a main argument the enhancement of reproductive autonomy. The ASRM also argued that it promotes social equality, although the cost of egg freezing may conversely be socially divisive. A key element of ethical practice involves the honest, accurate counselling of women about what the procedure involves both physically and emotionally, their individual expectation of success when contemplating this procedure, and its long-term implications, which may include decision-making about the fate of unused stored eggs. It is essential that this should include both the woman's age and centre-specific information. In some countries administrative regulations prohibit single women from accessing egg freezing, and legal challenges have been unsuccessful [19]. As a profession we need to consider thoughtfully our patient's desire for reproductive peace of mind and guard against a paternalistic judgement of their choices. Women may be doing this alone, hence feelings of isolation, anxiety and, in some cases, stigma may be heightened when seeking advice on egg freezing; regardless of whether or not they decide to proceed, supportive counselling may be required.

Evaluation of a patient decision aid for elective egg freezing in a randomised controlled trial in Australia has shown that this can reduce decisional conflict [20, 21]. No differences in distress or decisional regret were found between groups receiving or not receiving the decision aid at 12 months.

Marketing the technology as a form of 'reproductive insurance' is inappropriate, especially given the limited success rates in the women currently most likely to store eggs. Concerns include the use of social influencers to market egg freezing as paid partnership agreements.

Ethical issues also arise over eventual disposal of unused oocytes. The history of sperm cryostorage shows that this is problematic, but full discussion is beyond the remit of this brief paper.

5 | Safety of Freezing

While the greatest risks to children born following fertility treatment using frozen eggs are associated with multiple pregnancy and the sequelae of prematurity including cerebral palsy [22], there are no known additional risks specific to freezing such as congenital abnormalities [23]. Women who intend to use their own stored eggs at a later age should especially consider the risk of multiple pregnancy, which is related to age (as an index of egg quality) at storage. Multiple pregnancy should be minimised by use of a single embryo transfer strategy. Women embarking on pregnancy at a later age also experience greater obstetric risks, particularly in a first pregnancy, notably pre-eclampsia, gestational diabetes and the likelihood of a caesarean birth.

Storing a sufficient number of eggs requires the administration of gonadotrophins (injections of follicle stimulating hormone with or without luteinising hormone) to stimulate the growth of multiple ovarian follicles and therefore carries the risk of ovarian hyperstimulation syndrome (OHSS). The refinement of stimulation protocols (e.g., the use of a gonadotrophin-releasing hormone agonist to trigger ovulation instead of the standard human chorionic gonadotrophin) greatly reduces but does not avoid this risk. Possible complications from the egg harvest procedure include bleeding, pelvic infection, thrombosis and the risks of anaesthesia.

Long-term follow-up of children born from frozen eggs (including following vitrification) is needed; although the limited data available are reassuring. A systematic review [23], analysed data from 4519 babies to assess neonatal outcomes (multiple pregnancy, caesarean birth, gestational age at delivery, number of live births, birth weight, Apgar scores, congenital anomalies, baby health). However, the review presents little information beyond the neonatal period: data were available for only 245 babies at 12 months of age or later.

6 | Opinion

- Elective egg freezing provides women who are not in a position to start their family an opportunity to mitigate the inevitable decline in their fertility with increasing age.
- While often perceived (and promoted) as a form of insurance, it is essential that women undertaking egg freezing do so with a full understanding of the likelihood of success, as well as costs and risks.
- Success rates will be limited in women who are already in their mid–late 30s at the time of egg storage.
- An upper limit based on age (for both storage and use) might be more sensible for medical, biological and social reasons. We welcome the increased legal duration of storage; however, the lack of discrimination between elective and medically indicated egg storage by the HFEA prevents clear understanding of the changes in these two very different methods for female fertility preservation in the UK. We urge the regulator to find a way to solve this.
- The significant costs associated with the procedure and subsequent egg storage preclude many women from being able to consider it, raising issues of equality of access.
- Given that the NHS will not provide elective egg freezing, this service is provided in the private sector with inevitable commercial implications. With the recent introduction of state-funded elective egg freezing in France, other countries and insurance systems may reconsider their position.
- It seems likely that the future will see increasing numbers of women storing eggs, mostly because they are not in a relationship. There remains a need for societal changes that support women in the workplace to have their family at a biologically optimal age if they so choose, without compromising their career prospects, and with adequate provision for childcare that does not put those women/families at a financial disadvantage.
- There is a continued need to improve public education about age-related changes in female fertility, which should highlight the importance of men's knowledge as well as that of women.

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Declarations of Interest

Full declarations of interest are available upon request from the RCOG.

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